Topic: Introduction to Double-Ended Queue (Deque)

Q. What is a Double-Ended Queue (Deque)? Explain its types and how it differs from a standard queue and stack. Provide real-world applications.

> 📌 **Definition to Remember**
> A **Double-Ended Queue (Deque)** is a generalized linear data structure where insertion and deletion of elements can be performed at **both ends** (Front and Rear). It effectively combines the capabilities of both a Stack (LIFO) and a Queue (FIFO) into a single versatile structure.

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1. Operations in a Deque

Unlike standard queues (insert Rear, delete Front) or stacks (insert Top, delete Top), a Deque supports **four primary operations**:
1. Insert at Front
2. Insert at Rear
3. Delete at Front
4. Delete at Rear

Visual Representation:

```
Insert/Delete                          Insert/Delete
     ⟷ [ E1 | E2 | E3 | E4 ] ⟷
     Front                          Rear
```

2. Types of Restricted Deques

To enforce specific behaviors, Deques are classified into two restricted variants:

Type	Restriction	Allowed Operations
Input Restricted Deque	Insertion limited to ONE end	Insert (Rear only) Delete (Front & Rear)
Output Restricted Deque	Deletion limited to ONE end	Delete (Front only) Insert (Front & Rear)

3. Comparison with Stack and Queue

Data Structure	Insertion Point	Deletion Point	Principle
Stack	Top	Top	LIFO
Standard Queue	Rear	Front	FIFO
Deque	**Front & Rear**	**Front & Rear**	Both

4. Real-World Applications

- **Web Browser History:** When navigating back and forth, URLs are added/removed from one end. If history exceeds the max limit, oldest URLs are dropped from the other end.
- **Undo/Redo Operations:** Text editors use deques to maintain action histories.
- **Multiprocessor Scheduling (Work-Stealing):** If one processor finishes its own tasks, it can "steal" work from the rear of another busy processor's deque.
- **Palindrome Checking:** Reading a string simultaneously from the front and rear is efficiently handled by a Deque.

⭐ **Must-Write Points (for 10 marks)**

1. A Deque allows insertion and deletion at **both ends** (Front and Rear).
2. It combines the functionality of both Stacks and Queues.
3. Four main operations: Insert Front, Insert Rear, Delete Front, Delete Rear.
4. **Input Restricted Deque**: insertion at one end only, deletion at both.
5. **Output Restricted Deque**: deletion at one end only, insertion at both.
6. A standard queue restricts insertion to rear and deletion to front.
7. Key applications include Web Browser History, Undo/Redo, and the Work-Stealing scheduling algorithm.

⚡ **Quick Recall**

`Deque → Insert/Delete at BOTH ends → Input Restricted (Insert 1 end) → Output Restricted (Delete 1 end) → Apps: Browser History, Undo/Redo, Work-Stealing`